

Multi-Client Functional Encryption for Small Domains

Abstract. In this paper, we revisit the problem of multi-client functional encryption (MCFE) for general functions. Specifically, we consider the setting of secret-key MCFE for constant-arity functions where the input domain is polynomial in the security parameter. Surprisingly, we show that in this setting it is possible to construct a MCFE scheme secure for a bounded number of key and encryption queries based only on the minimal assumption that one-way functions exist. In contrast, all prior constructions of MCFE for general functions require very strong assumptions such as indistinguishability obfuscation or multi-linear maps.

Our main technique is to show that secret-key MCFE for polynomial input domain can be built from any secret-key multi-input functional encryption (MIFE) while inheriting the security properties of the underlying MIFE. Instantiating our construction with the MIFE of Brakerski et al. (Eurocrypt 2016) gives us a construction based only on the existence of one-way functions.

1 Introduction

Originating from a sequence of foundational works [1, 2, 3, 4, 5, 6, 7], functional encryption goes beyond the conventional “all-or-nothing” security guarantee of classical encryption. Specifically, functional encryption allows the issuance of a functional key corresponding to a function f that allows evaluation of the function f on encrypted data, while hiding everything else. Over the past decade, substantial effort has been devoted to studying both the security guarantees and the efficiency of functional encryption schemes [7, 6, 8, 9, 10, 11, 12, 13, 14].

While traditional, single-input, functional encryption has many applications it has a critical limitation – the entire input must be encrypted at one time into a single ciphertext. However, it is often desirable to evaluate functions over multiple ciphertexts for example if multiple pieces of data are encrypted over a long period of time, or if the encryptions are done by multiple different parties. To address this limitation, Goldwasser et al. [15] formalized two natural extensions: multi-input functional encryption (MIFE) and multi-client functional encryption (MCFE). MIFE allows the evaluation of multi-input functions over a set of ciphertexts, but all of these must be encrypted with the same encryption key, which must be kept secure. On the other hand, MCFE allows the input ciphertexts to be generated using different encryption keys, even allowing some of these keys to be controlled by an adversary. In both of these, the goal is to guarantee that nothing beyond the value of f on the encrypted inputs is revealed.

The early constructions of MCFE and MIFE for general functionalities, initiated by Goldwasser et al. [15], relied on the heavy cryptographic machinery of indistinguishability obfuscation [16, 13, 17] or multilinear maps [18]. Moreover, a further line of work showed that these assumptions are, in fact, necessary as unbounded variants of functional encryption imply obfuscation [19, 20, 21]. Thus, a fundamental question arises whether there is any version of MIFE and MCFE for general functions that can be realized from more standard, lighter assumptions.

Surprisingly, Brakerski et al. [22] showed that for the case of secret-key MIFE (i.e., a variant of MIFE where the master secret key is needed to encrypt) one can construct such a scheme for constant-arity functions that is secure against an adversary making a bounded number of key and encryption queries from the minimal assumption of one-way functions. However, this result did not give a construction of MCFE and, to date, the only constructions of secure MCFE are for restricted function classes such as inner products [23, 24, 25, 26, 27, 28, 29] and separable functions [30].

In this paper, we revisit the question of constructing MCFE from weaker assumptions and ask:

Is there a variant of MCFE for general functions that can be instantiated using only standard assumptions?

1.1 Our Contribution

We answer the above question in the affirmative. Our work departs from the prevailing paradigm for constructing MCFE schemes: Instead of limiting the class of admissible functions, we demonstrate that one can retain general functionality by restricting the input domain.

Specifically, we focus on the case of secret-key MCFE for all constant-arity functions computable by polynomial size circuits. In this setting, we show how to construct an MCFE scheme with polynomial size input domain that is secure against an adversary making a bounded number of key and encryption queries from *any* MIFE scheme. In particular, our construction only needs the MIFE scheme to be secure against an adversary making the same number of key queries and polynomially more encryption queries than the resulting MCFE scheme. Specifically, we prove the following theorems:

Theorem 1 (Informal). *Assuming the existence of a secret-key MIFE scheme for all constant-arity functions computable by polynomial size circuits, supporting a bounded number of encryption and key-generation queries, there exists a MCFE scheme for the same class of functions, defined over a polynomial-size input domain. The security of the resulting MCFE scheme reduces to that of the underlying MIFE scheme, incurring at most a polynomial overhead in the number of supported encryption queries.*

Instantiating the MIFE in the Theorem 1 with the secret-key MIFE construction by Brakerski et al. [22], we get the following corollary.

Corollary 1 (Informal). *Assuming the existence of one-way functions, there exists a MCFE scheme for all constant-arity functions computable by polynomial size circuits over polynomial size domain that is secure against an adversary making a bounded number of key and encryption queries.*

1.2 Our Approach

Observe that both MCFE and MIFE have essentially the same evaluation functionality. Given a functional key, and a set of ciphertexts, this procedure evaluates the function on the encrypted messages. Thus, an immediate idea is to somehow allow the MCFE clients to produce MIFE ciphertexts for their messages so that the MIFE evaluation procedure can be used. The main challenge in achieving this is that in MIFE all encryptions are generated under the same (secret) key, while in MCFE every client must receive its own encryption key, and moreover the scheme must remain secure even if some of these encryption keys are recovered by the adversary. It is thus not clear how one can use the MCFE encryption keys to generate ciphertexts under one MIFE key.

Our construction bypasses this challenge by leveraging the fact that the input domain is of polynomial size. Specifically, for a polynomial size domain, the trusted key authority can pre-encrypt MIFE ciphertexts for every possible message at every input slot. Since we are considering constant-arity functions, there are only polynomially many ciphertexts to generate. We then assign each client in MCFE to one of these input slots and give the corresponding ciphertexts to each client as their encryption key. With this key, a client can encrypt a message m by just choosing the m th ciphertext from its key. This allows the clients to generate MIFE ciphertexts for their messages without needing any information about the MIFE master secret key. Evaluation can then just use the MIFE key generation and evaluation procedure.

However, we are not quite done. The problem with the above construction is that the MCFE encryption procedure is deterministic, so each client can only encrypt each message exactly once. To deal with this limitation, we turn to the notion of labeled MCFE. In labeled MCFE each ciphertext additionally has a label and evaluation can only be done on ciphertexts with the same label thus preventing mix-and-match attacks where an adversary evaluates the function on all possible combinations of encrypted messages. To enable labels in our construction, we simply pre-generate ciphertexts for all message, label pairs – this is still polynomial if the number of possible labels is also small. Now, each client can encrypt a message for each label allowing a bounded number of encryptions. To ensure that labels are used correctly in evaluation, we modify the function in key generation queries to first check that all labels are equal before producing any output. This gives us a MCFE scheme able to answer a bounded number of encryption and keygen queries. Moreover, this scheme inherits its security properties from the underlying MIFE scheme, e.g., if the MIFE scheme is function-hiding, then so is the MCFE scheme.

Limitations Our construction does have one critical limitation. Like other labeled MCFE schemes, security only holds if clients never reuse a label for more than one encryption. In traditional MCFE [31, 32, 33] this is handled by using a timestamp or a hash output as the label, thus guaranteeing uniqueness. However, since our label space is small and must be fixed at initialization, we cannot use a randomly chosen label. Instead, we need the clients to be stateful to keep track of the labels they have already used, e.g., via a counter, or we need the application to pre-specify how clients can agree on a label from the pre-generated set in a deterministic way. We leave the question of how to remove this statefulness requirement as an interesting open question.

2 Applications

In this section we briefly discuss some potential application of our MCFE construction. We specifically focus on applications where small input domain and the bounded encryptions are a natural fit.

Electronic Voting : One natural application of our construction is for electronic voting where the final winner can be determined via MCFE evaluation of the client’s votes. In this setting, the input domain – i.e., the set of candidates – is inherently small and fixed (For future candidates, we can just use a number which will match the order candidates appear in the poll). Similarly, future election dates are predetermined and can naturally serve as labels in our framework, and it is easy for the clients to know the correct label at voting time. Moreover, the number of eligible voters is fixed in advance, which aligns directly with the arity of the underlying function.

Surveys: Surveys routinely ask respondents to choose from a small set of options or provide a ranking (e.g., from 1 to 10) for some item or activity. Our MCFE allows small-scale surveys to be carried out while maintaining the privacy of individual responses. In particular, our MCFE would enable evaluation of any statistics on the survey responses. Here the label for each survey can be the number of the survey, and surveys naturally have a small domain.

Multi-party Circuit PSI: Finally, our MCFE can be used to realize multi-party Circuit-PSI, e.g. [34] where the items in the sets come from a small domain, e.g., birthdays. Using our MCFE for this application gives a solution based only on secret-key assumption and with no interaction needed after the initial ciphertexts are sent. Furthermore, because the underlying MIFE scheme is proven secure against malicious evaluators, this security extends to the circuit PSI construction, offering strong protection against adversarial behavior during PSI evaluation.

3 Preliminaries

We summarize the notation and definitions used in this work.

Notation We let κ denote the security parameter. A function $\nu : \mathbb{N} \rightarrow \mathbb{N}$ is negligible if $\nu(\kappa) < \frac{1}{\kappa^c}$ for all $c > 0$ and sufficiently large κ . We use $[n] := \{1, \dots, n\}$. We denote t as the total number of clients, whereas $\mathcal{CP}$, $\mathcal{H}$ denote the set of corrupt and honest clients respectively st $|\mathcal{CP}| + |\mathcal{H}| = t$. We denote by q any polynomial st $q = q(\kappa)$. For a fixed constant $t > 1$, we let $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$ represent the set of all functions computable by polynomial size circuits where every $f \in \mathcal{F}_\kappa$ is of the form $f : \mathcal{X}_\kappa^t \rightarrow \mathcal{Y}_\kappa$ where $\mathcal{X}_\kappa^t, \mathcal{Y}_\kappa$ are the input space and output space respectively and $|\mathcal{X}_\kappa| < \text{poly}(\kappa)$.

$\mathbf{G}_{\text{MIFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa)$	$\mathbf{G}_{\text{MCFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa)$
INITIALIZE (1^κ) : 1 $b \leftarrow \{0, 1\}$ 2 $\text{msk} \leftarrow \text{Setup}(1^\kappa)$ ENCO $((x_{l,j}^0, x_{l,j}^1)_{l \in [t], j \in [q_2]})$: 3 $\forall l \in [t] : \forall j \in [q_2] :$ $\quad \text{ct}_{x_{l,j}}^b \leftarrow \text{Enc}_{\text{msk}}(x_{l,j}^b, l)$ 4 Return $(\text{ct}_{x_{l,j}}^b)_{l \in [t], j \in [q_2]}$ KEYGENO $((f_i^0, f_i^1)_{i \in [q_1]})$: 5 st $\forall j \in [q_2]$: $\quad f^0((x_{l,j}^b)_{l \in [t]}) = f^1((x_{l,j}^b)_{l \in [t]})$ 6 $\forall i \in [q_1] : \text{sk}_{f_i^b} \leftarrow \text{KG}_{\text{msk}}(f_i^b)_{i \in [q_1]}$ 7 Return $(\text{sk}_{f_i^b})_{i \in [q_1]}$ FINALIZE (b') : 8 Return $b' = b$	INITIALIZE $(1^\kappa, \mathcal{CP})$: 1 $b \leftarrow \{0, 1\}$ 2 $(\text{msk}, (\text{ek}_l)_{l \in [t]}) \leftarrow \text{Setup}(1^\kappa)$ 3 Return $(\text{ek}_l)_{l \in [\mathcal{CP}]}$ ENCO $((x_{l,j}^0, \text{lb}_{l,j}), (x_{l,j}^1, \text{lb}_{l,j}))_{l \in [\mathcal{H}], j \in [q_2]}$: 4 $\forall l \in [\mathcal{H}] : \forall j \in [q_2] :$ $\quad \text{ct}_{x_{l,j}, \text{lb}_{l,j}}^b \leftarrow \text{Enc}_{\text{ek}_l}(x_{l,j}^b, l, \text{lb}_{l,j})$ 5 Return $(\text{ct}_{x_{l,j}, \text{lb}_{l,j}}^b)_{l \in [\mathcal{H}], j \in [q_2]}$ KEYGENO $((f_i^0, f_i^1)_{i \in [q_1]})$: 6 st $\forall j \in [q_2] : \text{under } \text{lb}_j$ $\quad f^0((x_{l,j}^b)_{l \in [\mathcal{H}]}, \cdot) = f^1((x_{l,j}^b)_{l \in [\mathcal{H}]}, \cdot)$ 7 $\forall i \in [q_1] :$ $\quad \text{sk}_{f_i^b} \leftarrow \text{KG}_{\text{msk}}(f_i^b)_{i \in [q_1]}$ 8 Return $(\text{sk}_{f_i^b})_{i \in [q_1]}$ FINALIZE (b') : 9 Return $b' = b$

Fig. 1: Games defining Adaptive Message Security of MIFE (Left) and MCFE(Right)

3.1 Private Key Multi-Input Functional Encryption [35, 20]

For any $\kappa, t \in \mathbb{N}$, let $\mathcal{X} = \{\mathcal{X}_\kappa\}_{\kappa \in \mathbb{N}}$ be the plaintext space, $\mathcal{Y} = \{\mathcal{Y}_\kappa\}_{\kappa \in \mathbb{N}}$ be the output space and $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$ be the collection of all functions computable by polynomial size circuits where every f is of the form $f : \mathcal{X}_\kappa^t \rightarrow \mathcal{Y}_\kappa$. A private key t -input MIFE for $\mathcal{F}$ consists of quadruple of probabilistic polytime algorithms described as follows.

- **Setup** (1^κ) : On input the unary encoding of the security parameter κ , it outputs a master secret key msk
- **Enc** (msk, x, l) : On input an the master secret key msk , a message $x_l \in \mathcal{X}_\kappa$ for some slot $l \in t$, the encryption algorithm outputs a ciphertext ct_{x_l} .
- **KG** (msk, f) : On input a master secret key msk and a function $f \in \mathcal{F}_\kappa$, it outputs a functional key sk_f .

- $\text{Eval}(\text{sk}_f, (\text{ct}_{x_l})_{l \in t})$ takes as input a functional key sk_f and t ciphertexts $(\text{ct}_{x_l})_{l \in t}$, and outputs a string $y \in \mathcal{Y}_\kappa \cup \{\perp\}$.

Definition 1 (Correctness). A private key t input multi-input functional encryption scheme $\text{MIFE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Eval})$ for $\mathcal{F}_\kappa$ is correct if there exists a negligible function $\nu(\cdot)$ such that for every $\kappa \in \mathbb{N}$, for every $f \in \mathcal{F}_\kappa$, and for every $(x_l)_{l \in [t]} \in \mathcal{X}_\kappa^t$, it holds that

$$\Pr[\text{Eval}(\text{sk}_f, (\text{Enc}(\text{msk}, x_l, l))_{l \in t}) = f((x_l)_{l \in t})] \geq 1 - \nu(\kappa)$$

where $\text{msk} \leftarrow^* \text{Setup}(1^\kappa)$, $\text{sk}_f \leftarrow^* \text{KG}(\text{msk}, f)$, and the probability is taken over the internal random coins of Setup , Enc and KG .

In Figure 1(Left), we present the adaptive-message security game for MIFE [35, 15]. A stateful ppt adversary $\mathcal{A}$ interacts with a challenger that, upon receiving 1^κ , generates a master secret key and samples a hidden bit b . $\mathcal{A}$ issues q'_2 encryption queries to ENCO , receiving challenge ciphertexts, and q'_1 key-generation queries to KEYGENO , receiving functional keys f_i^b . Finally, $\mathcal{A}$ outputs a guess bit b' , and wins if $b' = b$. We now formalize a valid MIFE adversary

Definition 2 (Valid t -input adversary). A probabilistic polynomial-time algorithm $\mathcal{A}$ is a valid t -input adversary if for all private-key t -input multi-input functional encryption schemes $\text{MIFE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Eval})$ over a input space $\mathcal{X} = \{\mathcal{X}_\kappa^t\}$ and a function space $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$, for all $\kappa \in \mathbb{N}$ and $b \in \{0, 1\}$, and for all $(f^0, f^1) \in \mathcal{F}_\kappa$, $(x_l^0, x_l^1, l) \in \mathcal{X}_\kappa \times \mathcal{X}_\kappa \times [t]$ with which $\mathcal{A}$ queries the key-generation and encryption oracles, respectively, it holds that

$$f^0((x_{l,j}^0)_{l \in [t]}) = f^1((x_{l,j}^1)_{l \in [t]})$$

Definition 3 (Adaptive-message security). For all sufficiently large $\kappa \in \mathbb{N}$, a private-key t -input multi-input functional encryption scheme $\text{MIFE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Eval})$ over a message space $\mathcal{X} = \{\mathcal{X}_\kappa^t\}_{\kappa \in \mathbb{N}}$ and a function space $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$ with a q'_1, q'_2 bound on the number of KG and Enc queries respectively, is adaptive-message secure if for any valid t -input adversary $\mathcal{A}$ there exists a negligible function $\nu(\kappa)$ such that for MIFE as defined in Figure 1 the following holds

$$\text{Adv}_{\text{MIFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa) = \left| \Pr[\mathbf{G}_{\text{MIFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa) \Rightarrow 1] - \frac{1}{2} \right| \leq \nu(\kappa)$$

3.2 Multi Client Functional Encryption [15, 36]

For any $\kappa, t \in \mathbb{N}$, consider plaintext space $\mathcal{X} = \{\mathcal{X}_\kappa\}_{\kappa \in \mathbb{N}}$, label space $\mathcal{L} = \{\mathcal{L}_\kappa\}_{\kappa \in \mathbb{N}}$, output space $\mathcal{Y} = \{\mathcal{Y}_\kappa\}_{\kappa \in \mathbb{N}}$ and $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$ be the collection of all functions computable by polynomial size circuits where every $f \in \mathcal{F}_\kappa$ is of the form $f : \mathcal{X}_\kappa^t \rightarrow \mathcal{Y}_\kappa$. A private-key t -input multi-client functional encryption scheme MCFE for $\mathcal{F}$ consists of four ppt algorithms described as follows.

- $\text{Setup}(1^\kappa)$: On input the unary encoding of the security parameter κ , it outputs master secret key msk and t encryption keys $(\text{ek}_l)_{l \in [t]}$.

- $\text{Enc}(\text{ek}_l, x, l, \text{lb})$: On input an encryption key ek_l for some slot $l \in t$, a message $x \in \mathcal{X}_\kappa$, a label $\text{lb} \in \mathcal{L}_\kappa$, it outputs a ciphertext $\text{ct}_{x,\text{lb}}$.
- $\text{KG}(\text{msk}, f)$: On input a master secret key msk and a function $f \in \mathcal{F}_\kappa$, it outputs a functional key sk_f .
- $\text{Eval}(\text{sk}_f, (\text{ct}_{x_l, \text{lb}})_{l \in t})$ takes as input a functional key sk_f and t ciphertexts $(\text{ct}_{x_l, \text{lb}})_{l \in t}$, and outputs a string $y \in \mathcal{Y}_\kappa \cup \{\perp\}$.

Definition 4 (Correctness). A private key t input multi-client functional encryption scheme MCFE for $\mathcal{F}$ is correct if there exists a negligible function $\nu(\cdot)$ such that for every $\kappa \in \mathbb{N}$, for every $f \in \mathcal{F}_\kappa$, and for every $(x_l)_{l \in [t]} \in \mathcal{X}_\kappa^t$, it holds that

$$\Pr[\text{Eval}(\text{sk}_f, (\text{Enc}(\text{ek}_l, (x_l, l, \text{lb}))_{l \in t}) = f((x_l)_{l \in t})] \geq 1 - \nu(\kappa)$$

where $\text{msk}, (\text{ek}_l)_{l \in t} \leftarrow \text{Setup}(1^\kappa)$, $\text{sk}_f \leftarrow \text{KG}(\text{msk}, f)$, and the probability is taken over the internal random coins of Setup , Enc and KG .

In Figure 1(Right), we define adaptive-message security for MCFE [15, 36, 25]. A stateful ppt adversary $\mathcal{A}$ declares a corrupted set $\mathcal{CP}$, receives encryption keys for $\mathcal{CP}$, and may query encryption oracles on challenge messages for honest parties, using a common label across slots. The challenger samples a random bit b , encrypts messages, and returns ciphertexts; $\mathcal{A}$ may also query KEYGENO to obtain keys for functions f_i^b . Finally, $\mathcal{A}$ outputs a guess b' and wins if $b' = b$, and we formalize its admissibility and advantage below.

Definition 5 (Valid t -input adversary). A probabilistic polynomial-time algorithm $\mathcal{A}$ is a valid t -input adversary if for all private-key t -input multi-client functional encryption schemes $\text{MCFE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Eval})$ over a message space $\mathcal{X} = \{\mathcal{X}_\kappa\}_{\kappa \in \mathbb{N}}$ and a function space $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$, for all $\kappa \in \mathbb{N}$ and $b \in \{0, 1\}$, and for all $(f^0, f^1) \in \mathcal{F}_\kappa$, $\forall l \in \mathcal{H}$ where $\mathcal{H}$ are the honest parties, $(x_l^0, x_l^1, l, \text{lb}) \in \mathcal{X}_\kappa \times \mathcal{X}_\kappa \times [t] \times \mathcal{L}_\kappa$, with which $\mathcal{A}$ queries the key-generation and encryption oracles, respectively, and for all possible messages in the $\mathcal{CP}$ slots, such that for each query $\nexists l, l' \in [t], \text{lb}_l \neq \text{lb}_{l'}$, it holds that

$$f^0((x_{l,j}^0)_{l \in \mathcal{H}}, \cdot) = f^1((x_{l,j}^1)_{l \in \mathcal{H}}, \cdot)$$

The dot $(\cdot)$ above enforces that the above holds for all possible messages of the corrupt parties.

Definition 6 (Adaptive-message security). For all sufficiently large $\kappa \in \mathbb{N}$, a private-key t -input multi-client functional encryption scheme $\text{MCFE} = (\text{Setup}, \text{KG}, \text{Enc}, \text{Eval})$ over a message space $\mathcal{X} = \{\mathcal{X}_\kappa\}_{\kappa \in \mathbb{N}}$ and a function space $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$ with a q_1, q_2 bound on the number of KG and Enc queries respectively, is adaptive-message secure if for any valid t -input adversary $\mathcal{A}$ there exists a negligible function $\nu(\kappa)$ such that for MCFE as in Figure 1(Right) the following holds

$$\text{Adv}_{\text{MCFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa) = \left| \Pr[\mathbf{G}_{\text{MCFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa) \Rightarrow 1] - \frac{1}{2} \right| \leq \nu(\kappa)$$

4 Construction

In this section, we present our construction of an *adaptively* secure multi-client functional encryption scheme (MCFE) for constant-arity functions over *polynomial* size domains with bounded keygen and encryption oracle queries, built using a private-key multi-input functional encryption scheme (MIFE) for constant-arity functions as a black box. For a security parameter κ and a constant $t > 1$ representing number of parties, consider plaintext space $\mathcal{X} = \{\mathcal{X}_\kappa\}_{\kappa \in \mathbb{N}}$ where $|\mathcal{X}_\kappa| < \text{poly}(\kappa)$, label space $\mathcal{L} = \{\mathcal{L}_\kappa\}_{\kappa \in \mathbb{N}}$ such that $|\mathcal{L}_\kappa| < \text{poly}(\kappa)$ and output space $\mathcal{Y} = \{\mathcal{Y}_\kappa\}_{\kappa \in \mathbb{N}}$. Let $\mathcal{F} = \{\mathcal{F}_\kappa\}_{\kappa \in \mathbb{N}}$ be a set of all functions computable by polynomial size circuits where every $f \in \mathcal{F}_\kappa$ is of the form $f : \mathcal{X}_\kappa^t \rightarrow \mathcal{Y}_\kappa$. We provide below the construction of a MCFE scheme for $\mathcal{F}$.

4.1 Our Construction

Our approach, as outlined in Section 1.1, supports polynomial-size message and label spaces. Using the MIFE scheme, we start by generating the MIFE master secret key msk and use it to pre-compute ciphertexts for all plaintext and label using the MIFE encryption algorithm, we follow this approach for each slot $l \in [t]$. The generated table of ciphertexts for a slot $l \in [t]$ is then given to each of the t parties as the encryption key MCFE.ek_l for the MCFE scheme. To encrypt, each party $l \in [t]$, comes up with a message x from the message space $\mathcal{X}_\kappa$ and a label lb and uses the MCFE.ek_l to find the ciphertext corresponding to the tuple (x, lb) to get the corresponding MIFE ciphertext.

For any t -ary function f computable by $\text{poly}(\kappa)$ size circuit, the MCFE function key is simply the corresponding MIFE function key for f from $\mathcal{F}_\kappa$. Thereafter, an evaluator holding the functional key MCFE.sk_f can then compute the functional evaluation on the underlying plaintexts by calling the MCFE.Eval which in turn runs MIFE.Eval , provided that the ciphertexts supplied by all t parties were encrypted under a common label. This restriction is necessary to circumvent an inherent limitation of MIFE, namely the ability of an evaluator to arbitrarily *mix-and-match* ciphertexts from different time steps during evaluation.

As should be obvious from the above description, security of our scheme requires that each client only use each label in at most one ciphertext. This requires clients to ensure that they avoid label reuse. Since our label domain is small, we cannot achieve this by simply sampling labels at random. Thus, either the clients need to maintain a counter to indicate labels they have already used, or there needs to be some external mechanism that can assign labels from the pre-determined label space. We consider this mechanism to be external to the MCFE scheme and do not discuss it further.

Instantiation Instantiating our construction with the adaptive-message secure MIFE scheme by Brakerski et al.[22] which when further instantiated with the function-hiding single-input secret-key FE scheme of Gorbunov et al.[8], we obtain a bounded keygen and bounded encryption query MCFE for constrained domain by just relying on minimal assumption of one way function (OWF).

$\mathcal{D}_f((x_l, l, \text{lb}_l)_{l \in [t]}) :$ $\begin{array}{l} 1 \text{ if } \exists l, l' \in [t] \text{ st } l \neq l' \wedge \text{lb}_l \neq \text{lb}_{l'}: \text{Return } \perp \\ 2 \text{ Return } f(x_1, \dots, x_t) \end{array}$

MCFE.Setup(1^κ):

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1 MIFE.msk  $\leftarrow$  MIFE.Setup( $1^\kappa$ )
2  $\forall l \in [t]: sCT_l \leftarrow \emptyset$ 
3  $\forall m \in \mathcal{X}_\kappa:$ 
4    $\forall h \in \mathcal{L}_\kappa:$ 
5     MIFE.ctl,m,h  $\leftarrow$  MIFE.EncMIFE.msk(m, l, h)
6     sCTl = sCTl  $\cup$  {(m, h, MIFE.ctl,m,h)}
7 MCFE.ekl  $\leftarrow$  sCTl
8 MCFE.msk  $\leftarrow$  MIFE.msk
9 Return MCFE.msk, (MCFE.ekl)l ∈ [t]

```

MCFE.Enc(MCFE.ek_l, x, l, lb):

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10 parse {(m, h, MIFE.ctl,m,h): m ∈ Xκ, h ∈ Lκ}  $\leftarrow$  MCFE.ekl
11 MCFE.ctx,lb  $\leftarrow$  MIFE.ctl,m,h where m = x, h = lb
12 Return MCFE.ctx,lb

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MCFE.KG(MCFE.msk, f):

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13 MIFE.skD  $\leftarrow$  MIFE.KGMIFE.msk(Df)
14 Return MCFE.skf := MIFE.skD

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MCFE.Eval(MCFE.sk_f, (MCFE.ct_{x_l,lb_l)_{l ∈ [t]}):}

```

15 parse MIFE.skD  $\leftarrow$  MCFE.skf
16 parse MIFE.ctl,xl,lbl  $\leftarrow$  MCFE.ctxl,lbl
17 Return MIFE.EvalMIFE.skD((MIFE.ctl,xl,lbl)l ∈ [t])

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Fig. 2: MCFE Construction

Construction Details Let $\text{MIFE} = (\text{MIFE.Setup}, \text{MIFE.KG}, \text{MIFE.Enc}, \text{MIFE.Eval})$ be a secret-key multi-input functional encryption scheme for $\mathcal{F}$ that is also function hiding. We define MCFE as a quadruple of PPT algorithms $(\text{MCFE.Setup}, \text{MCFE.KG}, \text{MCFE.Enc}, \text{MCFE.Eval})$, each of which is briefly explained below. The formal construction is given in Figure 2.

MCFE.Setup(1^κ). On input the unary encoding of the security parameter κ , the Setup algorithm first invokes the setup procedure of MIFE to create the master secret key MIFE.msk . For each client slot $l \in [t]$, an initially empty ciphertext table is created. Then, for every plaintext $m \in \mathcal{X}_\kappa$ and label $h \in \mathcal{L}_\kappa$, the algorithm samples fresh randomness and applies the MIFE encryption procedure to generate the corresponding ciphertext. Each resulting ciphertext is stored in the table at index (m, h) . The completed table for slot l serves as the client's encryption key MCFE.ek_l . Finally, the algorithm outputs the MIFE master secret key together with the collection of all client encryption keys $(\text{MCFE.ek}_l)_{l \in [t]}$.

$\text{MCFE.Enc}(\text{MCFE.ek}_l, x, l, \text{lb})$. A client with an encryption key MCFE.ek_l , provides an input $x \in \mathcal{X}_\kappa$ and a label $\text{lb} \in \mathcal{L}_\kappa$ to the encryption algorithm which then retrieves the precomputed ciphertext for (x, lb) from the table MCFE.ek_l . This ciphertext is returned as the MCFE encryption of x for slot l under label lb .

$\text{MCFE.KG}(\text{MCFE.msk}, f)$. Functional key generation algorithm uses the MIFE.msk which is the same as MCFE.msk produced in **Setup** to run the MIFE key generation procedure on the input function $f \in \mathcal{F}_\kappa$. Thus invoking $\text{MIFE.KG}(\text{MIFE.msk}, \mathcal{D}_f)$ outputs $\text{MIFE.sk}_\mathcal{D}$ which we will use as MCFE.sk_f , where $\mathcal{D}_f$ is a functionality that simply ensures all inputs share the same label before evaluating f on the input plaintexts and returning the output. The function-hiding property of the underlying MIFE scheme hides f .

$\text{MCFE.Eval}(\text{MCFE.sk}_f, (\text{MCFE.ct}_{x_l, \text{lb}_l})_{l \in [t]})$. The evaluation algorithm takes as input the functional key MCFE.sk_f together with a ciphertext from each client. Recall that these ciphertexts are ciphertexts of the underlying MIFE scheme. Finally, the algorithm runs the MIFE evaluation procedure on these ciphertexts using MCFE.sk_f , to recover and thereby output $f(x_1, \dots, x_t)$ if all labels match, else $\perp$.

Proof of Correctness For all valid t messages $x_1, \dots, x_t$ under some label lb and for all function f from $\mathcal{F}_\kappa$, correctness of MCFE follows from the correctness of MIFE.

5 Proof of Security

The following theorem captures the security of the scheme stating that under appropriate assumptions on the underlying building block, the MCFE scheme is adaptive-message secure.

Theorem 1. *For any fixed integer $t > 1$ and security parameter κ , let MCFE for $\mathcal{F}$, as given in Figure 2, be defined over input space $\mathcal{X}_\kappa^t$ s.t. $|\mathcal{X}_\kappa| \leq \text{poly}(\kappa)$ and label space $\mathcal{L}_\kappa$ s.t. $|\mathcal{L}_\kappa| \leq \text{poly}(\kappa)$. Suppose MIFE is an adaptive-message secure multi-input functional encryption scheme for $\mathcal{F}$ with q'_1 and q'_2 bound on the number of KG and Enc queries respectively. Then MCFE is a adaptive-message secure multi-client functional encryption scheme for $\mathcal{F}$ with q_1 and q_2 bound on the number of KG and Enc queries respectively, where*

$$q_1 = q'_1 \quad \text{and} \quad q_2 = q'_2 - |\mathcal{X}_\kappa| \cdot |\mathcal{L}_\kappa|$$

And, for all $\kappa \in \mathbb{N}$, for any ppt adversary $\mathcal{A}$ against MCFE, there exists a ppt adversary $\mathcal{D}$ against MIFE such that

$$\text{Adv}_{\text{MCFE}, \mathcal{A}}^{\text{adp-ind}}(1^\kappa) \leq \text{Adv}_{\text{MIFE}, \mathcal{D}}^{\text{adp-ind}}(1^\kappa).$$

Proof. Towards proving Theorem 1, consider $\mathcal{A}$ to be a valid ppt adversary (as per Definition 5) against adaptively secure MCFE construction from Figure 2 making at most q_1 key-generation queries and q_2 encryption queries. Suppose $\mathcal{A}$ wins the experiment $\mathbf{G}_{\text{MCFE}, \mathcal{A}}^{\text{adp-ind}}$ in Figure 1(Right) with non-negligible probability. We construct another ppt adversary $\mathcal{D}$, valid as per Definition 2, that breaks the adaptive message security of MIFE in the experiment $\mathbf{G}_{\text{MIFE}, \mathcal{D}}^{\text{adp-ind}}$ of Figure 1(Left). Adversary $\mathcal{D}$ interacts with the MIFE challenger while internally simulating the MCFE game for $\mathcal{A}$.

In a nutshell, the proof begins with $\mathcal{A}$ submitting the set of corrupted parties, for which $\mathcal{D}$ must provide encryption keys comprising of ciphertexts for all plaintext-label pairs associated with those parties. Since $\mathcal{D}$ is a MIFE adversary, viewing parties as slots of the t -input function, it first queries its ENCO to generate ciphertexts for the corrupted slots, which are then returned to $\mathcal{A}$ as the corresponding encryption keys. Next, $\mathcal{A}$ submits challenge message queries for the honest parties. To address these, $\mathcal{D}$ embeds the honest queries into their corresponding slots, while filling the corrupted slots with dummy values, assigning identical messages to both challenge positions. The response from the challenger is then forwarded to $\mathcal{A}$. $\mathcal{A}$ may also issue functional key queries, which $\mathcal{D}$ translates directly into KEYGENO queries, with the corresponding responses returned to $\mathcal{A}$. It is easy to see that this is a perfect simulation of $\mathcal{A}$'s view in the MCFE security game. Thus, if $\mathcal{A}$ can distinguish between these ciphertexts, adversary $\mathcal{D}$ can break the security of the MIFE scheme. We provide details below.

Initialize: $\mathcal{A}$ begins the game by declaring the corrupt set of parties $\mathcal{CP}$ to $\mathcal{D}$. Now $\mathcal{D}$'s job is to generate the corrupt encryption keys. For this, $\mathcal{D}$ has to generate ciphertext for all possible plaintext-label pair for the corrupt set of parties by querying its challenge encryption oracle MIFE.ENCO. The number of queries that $\mathcal{D}$ needs to make to the MIFE.ENCO for simulating corrupt keys is $|\mathcal{X}_\kappa| \cdot |\mathcal{L}_\kappa|$.

Technically, to simulate the encryption keys for the corrupted parties, $\mathcal{D}$ maintains two lists M^0, M^1 each of size $|\mathcal{X}_\kappa| \cdot |\mathcal{L}_\kappa|$. For every message $m \in \mathcal{X}_\kappa$, and every label $\text{lb} \in \mathcal{L}_\kappa$, $\mathcal{D}$ creates a vector $(m, 1, \text{lb}), (m, 2, \text{lb}), \dots, (m, t, \text{lb})$ and adds it to both M^0 and M^1 . $\mathcal{D}$ then queries its encryption oracle MIFE.ENCO with M^0, M^1 and receives ciphertexts for each party for each message and label. It then uses the ciphertexts for the parties in $\mathcal{CP}$ as the corrupted ek 's and returns these to $\mathcal{A}$.

Encryption Oracle: After obtaining ek_l for each $l \in \mathcal{CP}$, $\mathcal{A}$ issues challenge queries for the honest parties of the form $\{(x_{l,j}^0, \text{lb}_{l,j})\}, \{(x_{l,j}^1, \text{lb}_{l,j})\}$ for $j \in [q_2]$ and $l \in [\mathcal{H}]$. To simulate these, $\mathcal{D}$ maintains two sets N^0, N^1 of size q_2 that serve as the challenge inputs for its challenge encryption oracle MIFE.ENCO. Each of these queries contain exactly t inputs, one for each slot. Since $\mathcal{D}$ has received the inputs only for $\mathcal{H}$ i.e Honest party, he needs to transform each query into MIFE.ENCO query. For each query $j \in [q_2]$, $\mathcal{D}$ does the following

- For each honest party $l \in \mathcal{H}$: insert $(x_{l,j}^0, l, \text{lb}_{l,j})$ into N^0 and $(x_{l,j}^1, l, \text{lb}_{l,j})$ into N^1 .

- For each corrupted party $l \in \mathcal{CP}$: choose any $m_j \in \mathcal{X}_\kappa$ and insert $(m_j, l, \text{lb}_{l,j})$ into both N^0 and N^1 .

D then queries its challenge encryption oracle MIFE.ENC0 with (N^0, N^1) and receives answers as MIFE.ct_2^σ where σ is the indistinguishability bit of its challenger. It extracts $\text{MCFE.ct}_{x_{l,j}, \text{lb}_{l,j}}^\sigma$ for each $l \in [\mathcal{H}], j \in [q_2]$, and forwards these to $\mathcal{A}$.

Let q'_2 be the total number of encryption oracle queries made to MIFE challenger by D, then from above we can say that

$$q'_2 \geq q_2 + |\mathcal{X}_\kappa| \cdot |\mathcal{L}_\kappa|$$

KeyGen Oracle: For a functional key query (f_i^0, f_i^1) from $\mathcal{A}$, consider $\mathcal{D}_f$ as the functionality that first checks consistency of labels and then evaluates f on the input plaintexts. Now for each such query, D queries $(\mathcal{D}_{f_i^0}, \mathcal{D}_{f_i^1})$ to its keygen oracle MIFE.KEYGEN0 . For its challenger's indistinguishability bit σ , it receives answer $\text{MIFE.sk}_\mathcal{D}^\sigma$, and provides it to $\mathcal{A}$ as the simulated MCFE functional key sk_{f_i} for each $i \in [q'_1]$. By construction, the admissibility checks in $\mathcal{D}$ ensure consistency with MCFE . The number of key queries is preserved, i.e.,

$$q'_1 = q_1.$$

Finalize: Eventually $\mathcal{A}$ outputs a bit b' . D outputs b' as its own guess for the challenge bit σ in the MIFE game. Clearly, if $\mathcal{A}$ distinguishes the challenge bit in the MCFE game with non-negligible advantage, then D does so in the MIFE game with the same advantage. Hence, the existence of $\mathcal{A}$ implies the existence of D breaking the adaptive-message security of MIFE , contradicting its assumed security.

Corollary 1. *For any security parameter κ , assuming one-way functions exist, there exists an adaptive-message secure secret-key MCFE scheme for $\mathcal{F}$ with query bounds q_1, q_2 on the number of KG and Enc queries, respectively.*

Proof. The proof of Corollary 1 follows by instantiating the MIFE in Theorem 1 with the secret-key MIFE scheme of Brakerski et al. [35], based on the secret-key single-input functional encryption scheme of Gorbunov et al. [8]. $\square$

Corollary 2. *For any fixed integer $t > 1$ and any security parameter κ , assume MIFE is a selective-message secure multi-input functional encryption scheme for $\mathcal{F}$ with q'_1 and q'_2 bound on the number of KG and Enc queries respectively, then there exists a selective-message secure multi-client functional encryption scheme for $\mathcal{F}$.*

Proof. The proof of Corollary 2 closely follows the approach in proof of Theorem 1, with the primary distinction that, in the selective-message security setting, all challenge encryption queries from the MCFE adversary are fixed apriori.

The remainder of the simulation and reduction argument proceeds analogously to the proof in Theorem 1.

□

6 Other Related Work

A large body of work has previously studied functional encryption in the multi-input and multi-client settings. We briefly review these papers here organized by the functions they support.

General FE: Research on multi-input general FE was started by Goldwasser et al. [15], who proposed a construction of MCFE / MIFE based on indistinguishability obfuscation. In terms of expressiveness, their scheme is remarkably powerful, supporting all multi-input functionalities that can be computed by bounded-size circuits. Building on this line of work, Boneh et al. [18] introduced a secret-key MIFE scheme with a stronger notion of security based on multilinear maps. While these papers achieved strong functionality, they both relied on strong assumptions, and moreover proved that these assumptions were necessary in some cases.

A key breakthrough came with the work of Ananth and Jain [20], who started the path towards weakening the assumptions by showing a selectively-secure secret-key MIFE scheme from general-purpose single-input functional encryption. Concurrently, Brakerski et al. [22] gave a secret-key MIFE scheme for constant-arity functions under weaker assumptions, including based only on the minimal assumption of one-way functions. However, both of these latter works only constructed MIFE leaving MCFE open for future work.

FE for Restricted Functionalities: Another line of work has instead looked at constructing MIFE/MCFE for restricted function classes. This line was started by Abdalla et al. [37] who showed how to construct MIFE for inner products from bilinear pairings. Further improvements by Abdalla et al. [38] developed a more general approach resulting in constructions from DDH, Paillier, and LWE. Later, Agrawal et al. [39] gave a construction of MIFE for quadratic functions.

The first construction of MCFE for inner products was given by Chotard et al. [32], proven secure under DDH in the random oracle model. Subsequent works obtained standard-model realizations: Abdalla et al. [23] under MDDH, DCR, and LWE assumptions, and Libert et al. [24] under LWE. Interestingly, Abdalla et al. [33] later demonstrated that their earlier MIFE construction [38] also gives an unlabeled MCFE scheme, thereby providing one of the first explicit connections between the two models. Beyond inner products, Ciampi et al. [30] constructed a MCFE scheme for separable functions.

Further work considered how to achieve function-privacy for MCFE. Agrawal et al. [39] presented the first function-hiding MCFE construction using bilinear pairings, proving its security in the random oracle model. Later, Shi and Vanjani [25] refined this approach to eliminate reliance on the random oracle model.

Notably, all known function-hiding MCFE schemes rely on pairings. In fact, a recent impossibility result by Ünal [40] shows that function-hiding IPFE schemes cannot be achieved from lattice-based assumptions, regardless of whether the setting is single-client or multi-client.

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